Bargaining over Cabinets: Multi-Agent RL for Portfolio Allocation in Coalition Governments

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GitHub: `https://github.com/bot2048/cabinet-bargaining-marl`

1 Problem and findings

After an election without a majority, parties bargain over who governs and how ministries are split. Real coalition cabinets divide portfolios almost exactly in proportion to seats (*Gamson's law*, slope ≈ 1 in $p = \alpha + \beta s + \gamma F$), whereas the leading bargaining theory, Baron–Ferejohn (BF), predicts that the *formateur* (the party proposing the cabinet) captures a large premium: the *portfolio allocation paradox*. We model parties as reinforcement-learning agents in a multi-agent environment that encodes government formation, rewarded only by their own office and policy payoffs (no fairness term), check them against the exact BF equilibrium, then switch institutional features on and off to find out which ones make the premium disappear. Findings:

F1 Validated learner. In the BF game, learned formateur shares match the closed form within 0.017 on average (10 seeds).

F2 Private information removes the formateur premium. Making parties' valuations private removes the formateur premium (from +0.145 to -0.010) and removing it from the full game restores the premium (+0.086): it is sufficient *and* necessary. Policy motivation is also sufficient.

F3 The premium moves into portfolio quality and shares follow power. With all mechanisms the formateur gets no extra ministries ($\tilde{\Phi} = -0.007$) but takes the ones it values most (premium +0.144 in its own valuation), and shares track Banzhaf voting power better than seats.

F4 Transfer to real parliaments. Without training on real data, agents placed in 242 real post-war Western European parliaments pick the governing coalition as often as classic size-based theories (21.7% vs 19.7%), and their allocations show no formateur premium.

2 RL formulation

Environment (`cabinet/env.py`): a partially observable stochastic game with n parties (seats w_i, majority quota q) and m ministries. Each period without a cabinet is a formation round: nature draws a formateur $f \sim \rho$, f proposes a winning coalition $C \ni f$ and an allocation $\mu : \{1..m\} \to C$, and the cabinet forms only if every partner accepts. A cabinet pays per-period flows until the term ends or a member walks out. Switchable mechanisms (`cabinet/config.py`): repeated terms with elections (M1), private types (M4), heterogeneous ministry values (M5), exits (M8), ministerial policy (M9), seat-proportional recognition (M10); level E0 is exactly the symmetric BF game.

State $\mathcal{S}$. Agent i observes a *set* of n party tokens (26 features: seat share, recognition probability, roles, pending/current shares, last Λ cabinets), m ministry tokens (8: salience, own valuation, policy dimension led), $m \times n$ pair features (4; partners' valuations and policy costs when types are public), a 17-dim context (phase, quota, δ, horizon, λ, ...) and 8 decision features. Tokens carry no labels: one network serves any n and m and is permutation-equivariant.

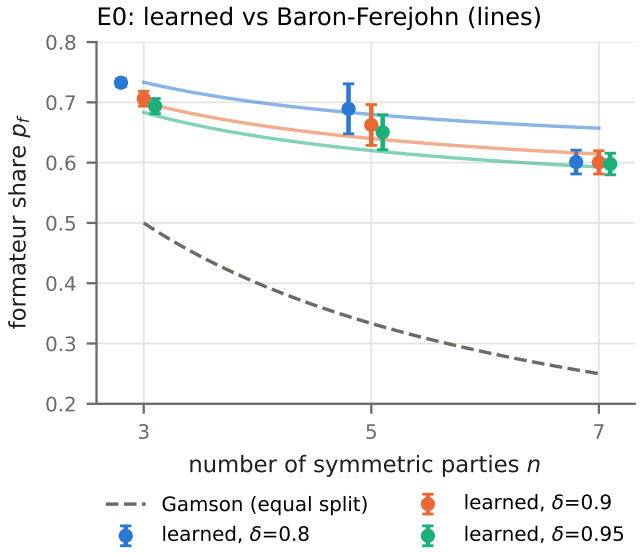


Figure 1: F1: learned formateur share (dots) vs the BF closed form (lines) in the symmetric game; dashed: Gamson's equal split.

Action $\mathcal{A}$ (masked, discrete). Formateur: include/exclude each other party (masks keep the coalition winning), then a pointer to a coalition member for each ministry; up to 10^{19} proposals are decoded in $n{-}1{+}m$ steps (a divisible surplus uses a 1% offer grid per partner). Voter: accept/reject. Governing member: stay/exit.

Reward $\mathcal{R}$. In a governed period member i receives $r_i = \sum_{k:\mu(k)=i} \nu_{ik} - \lambda \frac{1}{d} \sum_h |y_h - z_{ih}|$: the value of its ministries minus a policy loss (the minister holding the lead ministry of dimension h sets y_h). Outsiders get the policy term, an exiting member a bonus, failed rounds nothing; discount δ per period. Rewards are individual: a team reward would erase the conflict under study.

Algorithm (`cabinet/policy.py`, `cabinet/ppo.py`). One shared attention network with masked autoregressive proposal heads, vote/exit heads and a value head. Agents act only at some steps, so PPO with GAE runs on each agent's own decision sequence (semi-MDP). Learners: shared-parameter IPPO self-play (main), MAPPO with a centralised critic, fictitious self-play, PSRO; optional GRU memory. Baselines (`cabinet/baselines.py`): exact BF strategy from our solver (`cabinet/bf_solver.py`), a Gamson agent, a myopic BF heuristic and a random agent. Metrics (`cabinet/metrics.py`): formateur premium $\tilde{\Phi} = \mathbb{E}[p_f - s_f]$ and Gamson distance $\tilde{D}_G$, floor-adjusted for indivisible ministries, and regression (1). Scaled to one laptop: $n \leq 5$ (3–8 for real parliaments), $m = 6$, 6-period terms, 500–700 PPO iterations (4–6M decisions) per run, ≥ 3 seeds.

3 Results

F1, validation. Learned shares follow the BF closed form (Fig. 1; mean gap +0.001, 6/9 cells within 0.02), including its decrease with patience δ; errors grow with n, where each voter's threshold is learned less precisely. The same

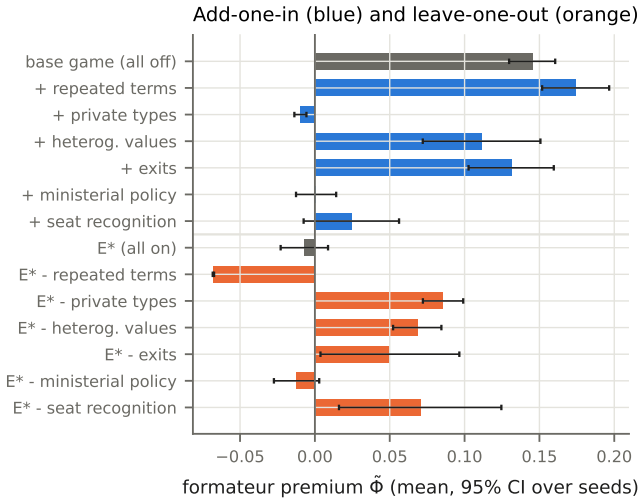


Figure 2: F2: formateur premium when one mechanism is added to the base game (blue: sufficiency) or removed from the full game E* (orange: necessity); mean and 95% CI over seeds.

Table 1: F3: the full game E* under different learners, memory and ministry counts.

	$\tilde{\Phi}$	$\tilde{D}_G$	$\hat{\beta}$	$\hat{\gamma}$	Φ_ν
IPPO, window Λ=4	-0.007	+0.093	+0.426	+0.054	+0.144
MAPPO (central critic)	-0.011	+0.085	+0.457	+0.038	+0.141
Fictitious self-play	-0.002	+0.097	+0.402	+0.063	+0.150
IPPO + GRU memory	–	–	–	–	–
IPPO, memoryless	-0.043	+0.098	+0.430	-0.008	+0.111
IPPO, 12 ministries	–	–	–	–	–

code then produces all other results, so differences between configurations come from the institutions, not the learner.

F2, what removes the premium. The base game keeps the BF premium ($\tilde{\Phi} = +0.145$, $\hat{\gamma} = +0.249$). Added alone (Fig. 2, blue), private types (-0.010) and ministerial policy ($+0.001$) each remove it and seat-proportional recognition nearly does ($+0.024$), while repetition, exits and heterogeneous values do not. Removed from E* (orange), private types bring it back ($+0.086$); removing values, exits or recognition restores part of it. The mechanism: when a formateur cannot see what partners value, it must over-offer to secure acceptance, which erodes its advantage. Repetition fails because agents discount the next term heavily (with patient agents: –; with GRU memory as well: –).

F3, where the premium goes. In E* the formateur no longer gets more ministries ($\tilde{\Phi} = -0.007$, $\hat{\gamma} = +0.054$) but its premium measured in its own valuation is $\Phi_\nu = +0.144$: it keeps the portfolios it cares about, a "quality" advantage that counts of ministries cannot see. Shares are flatter than seats ($\hat{\beta} = +0.426$): small partners are over-paid, the small-party bias of real cabinets, and voting power explains shares better than seats (R^2 0.56 for Banzhaf shares vs 0.44 for seat shares). This is what bargaining predicts (a pivotal small party is worth as much as a large one), so seat-proportionality itself does not emerge ($\tilde{D}_G = +0.093$). Table 1: these patterns hold across learners, memory and a finer 12-ministry game; symmetric PSRO gives –.

F4, real parliaments. Policies trained only on random parliaments were placed in the 242 post-war Western European majority coalitions of ParlGov, the real prime minister's party proposing (Fig. 3). They pick the governing coalition in 21.7% of cases, like Riker's minimal-seat rule (19.7%) and a random minimal winning coalition (21.7%), and 19.7% across 441 coalitions in 33 democracies. The ideologically closest coalition does better (31.7%): in this model a formateur can keep the policy-setting ministries

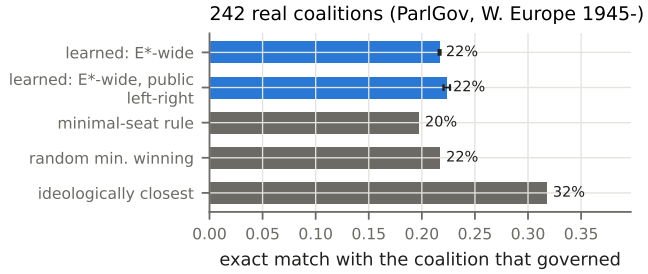


Figure 3: F4: coalition proposed by the real prime minister's party vs the coalition that governed.

itself, so partners' ideology barely matters to it, even when positions are visible (22.3%). On the real seat distributions the learned allocations show no formateur premium ($\tilde{\Phi} = -0.034$) and follow voting power (R^2 0.73 vs 0.64).

Reliability. Cross-play between independently trained runs costs – (same algorithm) and – (across algorithms) per period: runs reach compatible conventions. Best responses gain $+0.016$ per period against E* (– for MAPPO), near the 0.01 tolerance, but they also barely exploit a scripted Gamson profile ($+0.008$), so we claim approximate, not certified, equilibrium play.

4 Implications and limitations

Learning agents resolve one half of the portfolio paradox: the formateur premium of bargaining theory disappears once parties have private information (or care about policy), and what remains is a premium in portfolio *quality*, consistent with the empirical observation that weighting ministries by importance changes the picture. The other half, exact seat-proportionality, does not emerge from self-interested bargaining: learned shares follow voting power, which suggests that real cabinets' seat-proportionality reflects norms or pre-electoral bargaining that our agents lack. Limitations: scaled-down sizes; exploitability tests can refute but not prove equilibrium; the GRU is trained with stored states, without back-propagation through time; opponent shaping and the demand and pre-electoral modules of the specification were not run; ParlGov has no portfolio counts, so real allocations are compared through coalitions only. Coverage of the specification: `docs/SPEC_COVERAGE.md`.

Key code. `cabinet/env.py` (environment), `cabinet/config.py` (levels, mechanisms), `cabinet/policy.py` (policy), `cabinet/ppo.py` (PPO, MAPPO, FSP, exploitability), `cabinet/psro.py`, `cabinet/bf_solver.py`, `cabinet/metrics.py`, `scripts/run_experiment.py`, `scripts/parlgov_eval.py`, `tests/test_core.py`.

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